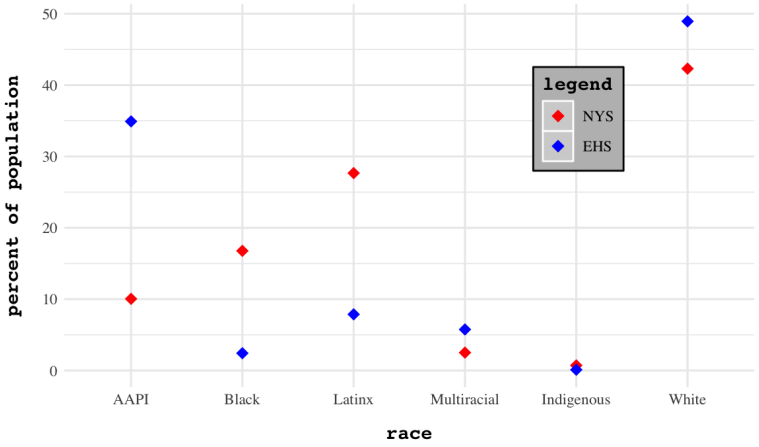
Edgemont Statistics

Statistics are all around us; nutrition, air quality, melting points, political opinions, weather, population—anything can be a statistic. There are two main types of data in statistics, categorical and numerical. Within each of these two categories are two sub-categories; nominal and ordinal for categorical data, then discrete and continuous for numerical data. You won't need to know what these subcategories are for our purposes, but feel free to look them up in your own time if you find yourself becoming interested in data science. Using data found online and the coding language "R," I analyzed some of EHS's statistics. I am by no means a data science expert; I'm just barely decent. This is not a groundbreaking scientific report that will expose the statistics of Edgemont, this is a sophomore's amateur analysis of readily available data. If you're interested, you can access my research, sources, and code in its entirety via the QR code at the end of the article.

My primary focus is going to be on race and ethnicity and how they affect different variables in education; comparing these effects between New York State as a whole and Edgemont specifically. Most of the data I found was from the 2020-2021 school year, which means that the data is not 100% accurate due to complications from COVID-19. Regardless, I hope my findings can help visualize Edgemont in comparison to the rest of the state.

The EHS student body is 51% BIPOC with an Asian majority. This is lower than the state average of about 59% with a Hispanic and African-American majority. *Figure 1* shows a simple comparison plot of the racial breakdowns in Edgemont and in NYS. One commonality we see in both NYS and EHS is the white population being the largest out of any racial group. The Indigenous student population locally and statewide are fairly similar in proportion. However, we must account for the fact that EHS only has 991 students, compared to the about 1.8 million students in grades 7-12 in New York. Edgemont has a higher white, multiracial, and AAPI population. New York State has a higher Indigenous, Latinx and African-American populations.

Figure 1 - racial breakdown



The most noticeable differences between NYS and EHS are the large disparities between their AAPI, Black, and Latinx populations. There are far more AAPI students in EHS than in the average New York public school. Later figures will reveal to us that the AAPI population in Edgemont is mostly Asian, and not Pacific Islander. Many more observations can be made from this graph—but this is a school newspaper article, not a scientific report—so let's stop here for now. Moving onto *Figure 2*, we see the ethnic distributions in Edgemont for the past two decades. The populations of most racial groups are climbing steadily, except for the white population and the Indigenous population. The number of white students in Edgemont over time has decreased from 2006 to 2012 when it starts to plateau. The number of Native American students, however, remains at almost 0 for the past two decades. Previously, we mentioned that

Figure 2 - EHS racial distribution timeline

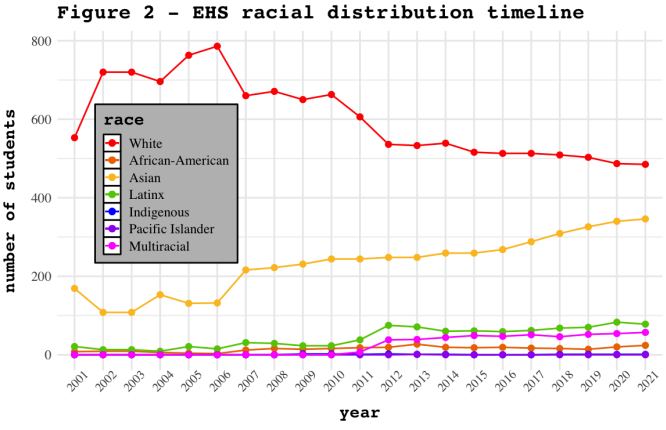
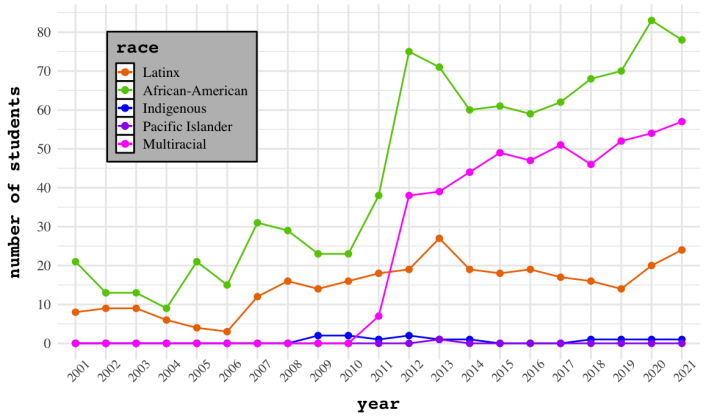


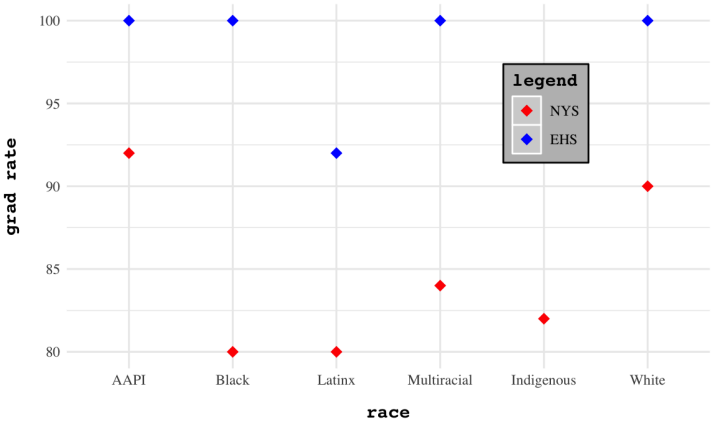
figure 3 - minority populations



Edgemont had a higher AAPI population than the typical New York high school. When we look closer at what makes up the EHS AAPI student body, we see that it's mostly—if not entirely—Asian Americans. *Figure 2* shows that, similarly to EHS' Indigenous population over the years, there are little to no Pacific Islander students at Edgemont. At first glance, we see that Asians and whites are the majority, *Figure 3* gives us a closer look at the racial distribution of the minority population.

Edgemont's graduation rates are in the top 10% of all schools in New York State. Students at EHS are well educated and are among the top 20% in overall test scores; so it's only natural that we would have a high graduation rate. *Figure 4* displays the graduation rates of both EHS and NYS. As we can see, Edgemont has a phenomenal graduation rate. The one outlier of 92% can be explained by the fact that there were only 13 Latinx students in the 4-year August 2017 cohort; and one of them, instead of graduating, was still enrolled. Edgemont had no dropouts in this cohort according to data.nysed.gov, one of my main sources for this data. Looking at the graduation rate for Indigenous students in Edgemont, we see that there is actually no point plotted. This is because there were zero Indigenous students in the cohort. This lines up with the data in Figures *3* and *4*, which show a small-to-nonexistent Native American population. The average graduation rate for Indigenous students in New York is 84%. Edgemont's proficiency in math and reading is 88% and 83% respectively. This compared to the state average math and reading proficiency of 54% and 52% further shows how well-educat

Figure 4 - grad rates for various ethnicities



ed students in EHS are.

New York State's average diversity score is 0.73, while 0.69 is the national average. As of the 2020-2021 school year, Edgemont's diversity score is 0.63—lower in comparison to both the national and state averages. Diversity score is calculated with Simpson's Diversity Index formula, which determines the probablility of two randomly selected entities being of different subgroups. The formula is $D = 1 - (\frac{\Sigma n(n-1)}{N(N-1)})$, *N* represents the total population and *n* represents the subgroup population. Using this simple formula, I calculated the EHS diversity scores of the past two decades and plotted them in *Figure 5*. I've included some annotations to better visualize the data. The diversity of Edgemont has been increasing over time, but we have reached a plateau in the past 7 or so years. Will this plateau lead to an increase similar to the one after the plateau in 2007-2010? We'll have to wait to find out!

figure 5 - diversity score timeline

